

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. (5 Marks)

Understanding of Data

The RL agent is trained over multiple episodes, and its reward increases gradually from 10 to 45. This means the agent is receiving increasingly better outcomes as training progresses.

Analysis

The upward trend in reward indicates that the agent is learning from its previous interactions with the environment. Initially, the agent may take random or poor actions, resulting in low rewards. As the number of episodes increases, it learns which actions produce better results.

The improvement is:

Reward increase = $45 - 10 = 35$

This represents a significant improvement in the agent's performance.

Application of Concepts

In Reinforcement Learning, the agent updates its **policy or value estimates** using the rewards received from the environment. Higher rewards over time indicate that the agent is selecting better actions for different states.

A gradually increasing reward generally suggests that the learned policy is becoming more effective.

Conclusion

The increase from 10 to 45 indicates positive learning performance and policy improvement. However, an optimal policy can only be confidently claimed if the reward eventually stabilizes near a high value over many additional episodes.

Final Answer

The increasing reward trend shows that the agent is learning successfully and improving its policy.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. (5 Marks)

Understanding of Data

The agent is tested with three epsilon values:

- $\epsilon = 0.1 \rightarrow$ Highest cumulative reward
- $\epsilon = 0.3 \rightarrow$ Intermediate reward
- $\epsilon = 0.5 \rightarrow$ Lowest cumulative reward

Here, ϵ controls the probability of **exploration**.

Analysis

In an ϵ -greedy strategy:

- Exploration: Try a random/new action.
- Exploitation: Select the currently best-known action.

Therefore:

Epsilon Exploration Exploitation Observed Performance

0.1	10%	90%	Highest
0.3	30%	70%	Medium
0.5	50%	50%	Lowest

With $\epsilon = 0.5$, the agent explores half of the time. This can cause it to select less rewarding actions more frequently.

Application of Concepts

The results show that exploitation is more beneficial in this particular trained environment. Since the agent already has useful knowledge, choosing the best-known action more frequently produces higher cumulative reward.

Conclusion

The best-performing value is:

$$\epsilon = 0.1$$

because it provides 90% exploitation and only 10% exploration, resulting in the highest cumulative reward.

Final Answer

$\epsilon = 0.1$ provides better performance based on the observed data because it balances limited exploration with strong exploitation of learned knowledge.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)

Understanding of Data

Two RL algorithms, Q-Learning and SARSA, are compared at 100, 200, and 300 episodes. Q-Learning consistently obtains slightly higher average rewards than SARSA.

Analysis

The observed pattern is:

Q-Learning reward > SARSA reward

at 100, 200, and 300 episodes.

This means Q-Learning is achieving slightly better performance under the given experimental conditions.

Application of Concepts

The main difference is:

- Q-Learning: An off-policy algorithm. It learns the value of the best possible next action.
- SARSA: An on-policy algorithm. It learns using the action that the current policy actually chooses.

Q-Learning therefore tends to learn a more aggressive optimal policy, while SARSA considers the behavior of the current policy, including exploration.

Analysis of Convergence

Since Q-Learning has consistently higher rewards at all three checkpoints, the observed data suggests that Q-Learning is learning more effectively under these conditions.

However, higher reward alone does not mathematically prove faster convergence. To establish convergence speed, we would ideally need to see which algorithm reaches a stable reward level earlier.

Conclusion

Based on the observed results, Q-Learning performs better than SARSA because it consistently achieves higher average rewards. If the higher reward also stabilizes earlier, Q-Learning can be considered to converge faster.

Final Answer

Q-Learning shows better observed performance. The data supports better learning performance, but convergence speed should be confirmed from reward stabilization rather than reward magnitude alone.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. (5 Marks)

Understanding of Data

The agent is trained using discount factors:

- $\gamma = 0.2$
- $\gamma = 0.5$
- $\gamma = 0.9$

The average reward increases as the discount factor increases.

Analysis

The discount factor γ determines how much the agent values future rewards.

The Bellman value equation is:

$$V(s) = R + \gamma V(s')$$

where γ controls the importance of future rewards.

- $\gamma = 0.2$: Strong focus on immediate rewards.
- $\gamma = 0.5$: Gives moderate importance to future rewards.

- $\gamma = 0.9$: Gives high importance to future rewards.

Since the average reward increases as γ increases, the environment appears to benefit from long-term planning.

Application of Concepts

With $\gamma = 0.9$, the agent considers future consequences more strongly. Therefore, it may choose an action that gives a smaller immediate reward but produces larger rewards later.

For example:

Immediate reward = 5
Future reward = 20

A high γ makes the future reward more valuable to the agent.

Conclusion

Since $\gamma = 0.9$ produces the highest average reward, it is the most suitable value among the tested values for this problem.

However, a high discount factor is not universally best; it depends on the nature of the environment.

Final Answer

$\gamma = 0.9$ is most suitable based on the given data because it emphasizes long-term rewards and produces the highest average reward.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. (5 Marks)

Understanding of Data

The success rate increases from:

45% at 50 episodes → 92% at 200 episodes

The improvement is:

$$92\% - 45\% = 47\%$$

So, the agent's success rate improves by 47 percentage points.

Analysis

At 50 episodes, the agent succeeds only 45% of the time, indicating that its policy is still developing.

At 200 episodes, the success rate reaches 92%, showing that the agent has learned to select effective actions much more frequently.

The trend is strongly positive:

45% → 92%

This indicates substantial learning during training.

Application of Concepts

In RL, the agent learns by repeatedly interacting with the environment. Successful actions receive positive rewards, and the agent updates its policy/value estimates.

As training episodes increase, the agent becomes better at identifying actions that lead to successful outcomes.

Conclusion

The increase to 92% success rate strongly indicates that the agent has learned a high-performing policy.

However, 92% alone does not prove that the policy is mathematically optimal. To claim an optimal policy, we should also verify that the success rate remains stable over additional episodes and is close to the best possible performance.

Final Answer

The increase from 45% to 92% demonstrates significant learning and indicates that the agent has developed a highly effective policy. Further stable performance would be needed to confirm optimality.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand